ASTRA P1 RESEARCH CONTINUATION

Early information as a description of the remaining experiment

Exact control-value theory, a finite polynomial construction,
and a representation-discovery boundary

30 October 2026 UTC

Scientific objective. Predict a task accurately while paying for the measurements, the description needed to choose them, model acquisition, and local computation. The central question is when early information makes the remaining experiment easy to describe, even though the target has no useful small fixed linear representation.

Strongest completed result. For a conditional-linear target with an independent Gaussian residual input, every interactive finite-control sensing protocol has exactly the value of a static finite codebook of measurement precision matrices. A Hermite construction shows why the distribution of the coefficient description matters: covariance can preserve every fixed-experiment risk and still miss a polynomial control separation.

Concrete accuracy result. A normalized cubic family with 8,384 Gaussian input coordinates admits a 129-read noisy protocol with final MSE below 0.03, including finite context messages, finite local actuator weights, and a finite final ADC. Every fixed 129-read linear experiment, even mixing all coordinates and using an arbitrary decoder, has MSE at least 0.953589.

Operational boundary. The protocol generates and applies 8,256 actuator weights. Its short incoming message does not remove this work. A native direct-target evaluator implements the same conditional operation. No physical sensor, generic structure learner, runtime advantage, or historical first is established.

Ordinary proofs, executable checks, acquired model, cost ledgers, independent audits, corrections, and failed transfers accompany this report. Earlier P1 artifacts are preserved. Capital integration is separate.

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1 The opening and what changed

The capability sought is accurate prediction from an expensive, distributed, or partially observed state. A useful first experiment should do more than remove uncertainty: it should make the remaining task inexpensive to describe and implement. This is an organizing question about information and representation, rather than an assumption that a large adaptive policy must be searched.

The previous campaign established a two-round comparison for known Gaussian quadratic targets, an optimal-fixed noisy separation, an exact inertia classification, coordinate companions, and paid coefficient acquisition. Its final scalar read could require a dense control direction. The present construction examines that resource directly and then replaces the unstructured direction by a nonlinear family with a small generating description.

Three findings form the main result:

1. **An exact reduction of feedback to description.** In a stated conditional-linear Gaussian interface, arbitrary interaction has the same optimum as choosing one matrix from a finite codebook. This includes noisy, repeated and nonorthogonal measurements.
2. **A separation that covariance cannot express.** A coefficient vector can have isotropic covariance in M dimensions while being generated by $r \ll M$ variables. A constructive example preserves final noisy accuracy using $O(r)$ incoming control bits and $r+1$ reads, while every fixed linear experiment needs $\Omega(M)$ reads for constant accuracy.
3. **A boundary on automatic discovery.** Recognizing even one global affine direction of an arbitrary supplied cubic is NP-hard, with an inverse-polynomial gap. The easy layout in the constructive family is therefore a substantive promise; learning its coefficients does not discover an arbitrary factorization.

The conditional-linear representation belongs to the established varying-coefficient framework of Hastie and Tibshirani [1]. Gaussian conditioning, Hermite polynomials, quantization and aligned target evaluation are also established machinery. The combination is proved here without a historical-priority claim. Jackowski's average-case multilinear work remains a close unresolved comparison [5].

2 A precise access contract

Let $C \in \mathbb{R}^m$ have an arbitrary known square-integrable law, independently of $V \sim N(0, I_m)$. The target is $F = C^\top V$. The controller knows C. The actuator may take at most q reads

$$Y_t = a_t^\top V + \xi_t, \qquad \|a_t\| \le 1, \qquad \xi_t \overset{\text{iid}}{\sim} N(0, s), \quad s > 0.$$

All physical noises and algorithmic seeds are independent of (C, V). The actuator selects directions from its received control, past readouts and seeds. The controller may receive the readouts and interleave control bits with measurements. The *final predictor retains* C and receives the legal transcript.

The hard control budget is a binary protocol with at most B controller branches on every path. Every coefficient-dependent distinction at the actuator—including timing, requests, addresses, stopping and metadata—is charged. Thus every counterfactual replay has at most $K = 2^B$ terminal branches. Each branch is totalized as a legal protocol with at most q reads. Free, unframed termination is not an extra communication alphabet.

The theorem prices outgoing controller bits and real scalar reads. Acquiring C, computing conditional means, returning readouts, programming directions, codebook storage, and local arithmetic remain separate resources. The later construction pays specified finite messages and patterns, rather than treating these additional operations as free.

3 Exact value of arbitrary noisy interaction

For a fixed matrix M of at most q rows with norm at most one, write

$$\Sigma_s(M) = \left(I + \frac{M^\top M}{s}\right)^{-1}.$$

An optimal design may use q unit rows: appending a read or increasing a row's power cannot worsen its information. Repeated rows add precision and are permitted.

Theorem 1 (Finite control equals a static design codebook)**.** *Under the preceding contract, the minimum Bayes squared error over all interactive protocols is*

$$\boxed{\mathcal{R}_s(B, q; C) = \min_{M_1,\dots,M_{2^B}} \mathbb{E}_C \min_j C^\top \Sigma_s(M_j) C.} \tag{1}$$

The matrices range over fixed feasible sensing designs. One message selecting one design attains this optimum.

Proof. First condition on C, the legal transcript and all algorithmic seeds. Do *not* condition on the physical noise innovations. The realized directions are predictable from past observations. Sequential Gaussian likelihoods give

$$\mathrm{Cov}(V \mid C, \text{transcript}, \text{seeds}) = \Sigma_s(M_{\text{actual}}).$$

Observed messages, choices and stopping add only consistency conditions determined by the fixed coefficient and numerical history. The conditional Bayes variance is therefore $C^\top \Sigma_s(M_{\text{actual}}) C$. If any seeds were hidden from the decoder, granting them only strengthens this lower bound.

Now, *after deriving that variance expression*, freeze a full value of V, the pre-sampled ordered noise innovations, and all seeds. Replay the actuator for every possible control branch, supplying $a_t^\top V + \xi_t$ at its t-th requested read. At a controller decision, branch on the bit instead of invoking the encoder. The resulting collection $\{M_j\}$ contains at most 2^B feasible matrices and is independent of C. The actual trajectory is one of them. Pointwise,

$$C^\top \Sigma_s(M_{\text{actual}}) C \geq \min_j C^\top \Sigma_s(M_j) C.$$

For each frozen replay collection, independence allows averaging over C. Its expected minimum is at least the optimum over all static codebooks in (1). Averaging over the frozen quantities proves the lower bound. Freezing the noise is a proof coupling, not information given to the decoder.

Conversely, choose an optimal codebook and send the index minimizing its quadratic form for the observed C. Execute that matrix and use the ordinary Gaussian posterior mean of $C^\top V$. Its risk is the right side of (1). Existence follows from compactness of the feasible matrices and dominated convergence, with dominating variable $\|C\|^2$. □

This removes the need to construct a comparator adaptive policy in the proof. It does *not* give an efficient optimizer for a 2^B-entry codebook. Exponential storage or exhaustive selection can erase any operational value. Efficient structure must be earned separately.

3.1 Exact reads and endpoint checks

For exact observations, the corresponding identity is

$$\mathcal{R}_0(B, q; C) = \min_{\dim S_j \le q} \mathbb{E} \min_j \left\| (I - P_{S_j})C \right\|^2. \tag{2}$$

It follows directly from Gaussian innovations, or by bounding noisy designs below by their projection residuals and above by orthonormal implementations as $s \downarrow 0$. In particular, with one noisy read,

$$\mathcal{R}_s(B, 1; C) = \frac{s}{1+s}\mathbb{E}\,\|C\|^2 + \frac{1}{1+s}\mathcal{R}_0(B, 1; C).$$

With unlimited control, q repeats along C attain the exact value $s\mathbb{E}\,\|C\|^2/(s+q)$. With zero control, $C \sim N(0, I_m/m)$ and $q \le m$, orthonormal reads attain $1 - q/[m(1+s)]$. These checks retain noisy measurement strength rather than collapsing repeated reads into an unweighted subspace.

3.2 Necessary access corrections

If the actuator must predict without retaining C, the theorem changes. For $m = q = 1$, $B = 0$, and C an independent random sign, its best prediction of CV remains zero even if it learns V exactly. A controller retaining C has zero exact-read error. Output location is mathematically consequential.

If public actuator context Z is correlated with C, while V remains independent of (C, Z), the correct value is

$$\mathbb{E}_Z \mathcal{R}_s(B, q; \mathcal{L}(C \mid Z)).$$

Public first-stage randomness can become correlated with the eventual posterior coefficient and must be handled this way. If the actuator already retains C locally, the external B-bit restriction is absent; its local computation and programming costs remain.

Neither the theorem nor its proof extends without further work to a nonlinear target in the remaining V, non-Gaussian V, arbitrary dependent noise, uncharged side information, or an expected read budget.

4 What an early experiment changes

Let X be independent of $V \sim N(0, I_m)$, and consider

$$F = g(X) + a(X)^\top V, \qquad \mathbb{E}g(X)^2 + \mathbb{E}\,\|a(X)\|^2 < \infty.$$

The first experiment H observes X only, with randomness independent of V. The controller subsequently obtains no additional X information. Define

$$\bar{g}(H) = \mathbb{E}[g(X) \mid H], \qquad C(H) = \mathbb{E}[a(X) \mid H],$$
$$J_0 = \mathbb{E}\,\mathrm{Var}(g(X) \mid H) + \mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}(a(X) \mid H).$$

Theorem 2 (Uncertainty plus description cost)**.** *When the final predictor retains H, the exact optimal continuation risk is*

$$J_0 + \mathcal{R}_s(B, q; C(H)), \tag{3}$$

with the preceding conditional-codebook qualification if the actuator has correlated public context.

Proof. Decompose

$$F = \bar{g}(H) + C(H)^\top V + E_0, \quad E_0 = g - \bar{g} + (a - C)^\top V.$$

Independence gives $\mathbb{E}[E_0 \mid H, V] = 0$. Later legal transcripts are generated from H, V and independent noises and seeds, so E_0 is orthogonal to every resulting predictor. Averaging its squared norm gives J_0, because $\mathbb{E}V = 0$ and $\mathbb{E}VV^\top = I$. Thus the squared error splits exactly into J_0 plus the error on $\bar{g} + C^\top V$. Subtract the known $\bar{g}$ and apply the previous theorem. Extra controller information in H beyond $C(H)$ cannot improve the replay lower bound. □

Conditional covariance between g and a need not vanish. Its cross term cancels after averaging V; J_0 is not generally a constant residual variance at every terminal transcript. Also, evaluating $C(H)$ or $\bar{g}(H)$ may itself be the hard inference problem. Equation (3) supplies an exact objective, not a conditional-mean oracle.

This identifies two effects of early information. It reduces irreducible uncertainty J_0, and it changes the distribution of descriptions needed for the remaining experiment. Optimizing only the first effect can miss the second.

5 Why covariance is insufficient

Suppose D is any fixed observation of V, independent of C. Even when D is nonlinear,

$$\mathbb{E}\,\mathrm{Var}(C^\top V \mid C, D) = \mathrm{tr}\Big(\mathbb{E}[CC^\top]\,\mathbb{E}[\mathrm{Cov}(V \mid D)]\Big). \tag{4}$$

This follows by conditioning on C, D and then using independence. Two coefficient laws with the same second-moment matrix therefore have the same risk for every such fixed experiment. This is a fixed-V-experiment statement with coefficient knowledge at the decoder, not a claim about arbitrary joint measurements of the source of C and V.

Now compare $C_G \sim N(0, I_M/M)$ with

$$C_H = \phi(U)/\sqrt{M}, \qquad U \sim N(0, I_r), \quad M = r(r+1)/2,$$

where ϕ consists of the orthonormal degree-two Hermite features. Both coefficient laws have mean zero and covariance I_M/M. The first requires $\Omega(M)$ control bits for a constant-error single exact read. The second has an explicit $O(r)$-bit control representation at fixed error. The same covariance can therefore preserve every fixed risk while destroying the distinction relevant to controlled experiments.

5.1 The unstructured Gaussian obstruction is established quantization theory

For C_G and $1 \le q < M$, let $d = M - q$, $a = d/2$, $b = q/2$, $K = 2^B$, and let $F_{a,b}$ be the beta CDF. The projection distortion has bounds

$$\int_0^1 (1 - KF_{a,b}(x))_+ \, dx \le \mathcal{R}_0(B, q; C_G) \le \int_0^1 (1 - F_{a,b}(x))^K \, dx. \tag{5}$$

The lower bound is a union bound over subspaces; the upper is the mean distortion of a random Haar codebook. It is an existence result, not a cheap encoder. For fixed dimensions at high rate, distortion scales as $K^{-2/(M-q)}$ up to constants. This unequal-dimensional Grassmann quantization

and its rate-distortion theory are prior work [2]. Structured quantizers such as Cube-Split are relevant efficient alternatives to explicit exponential search [3].

For an expected prefix length at most B, a separately proved exact-read obstruction is

$$B \log 2 \geq \frac{d}{2}\left[\log \frac{d}{MR} - 1 + \frac{MR}{d}\right], \quad 0 < R < d/M. \tag{6}$$

For $R \geq d/M$, this argument gives zero. To prove it, condition on the full replay input and independent seeds. Prefix coding bounds the conditional mutual information by $B \log 2$. Under the product reference law, a residual projection has at least d Gaussian degrees of freedom, so its negative quadratic Laplace transform is bounded by $(1+2t)^{-d/2}$. The variational entropy inequality gives $(d/2)\log(1+2t) - tMR$; optimize over $t \geq 0$. This does not extend the exact hard-budget identity to all expected-budget problems.

6 A polynomial family with a short nonlinear description

Let $p \geq 1$, let ϕ_γ be the normalized degree-p multivariate Hermite basis, and put

$$M = \binom{r+p-1}{p}, \quad F_c(U,V) = \frac{1}{\sqrt{M}} \sum_{|\gamma|=p} c_\gamma \phi_\gamma(U) V_\gamma,$$

where $U \sim N(0, I_r)$ and $V \sim N(0, I_M)$ are independent. Its variance is $\sigma_c^2 = \|c\|^2 / M$. For $c_\gamma = 1$ this is one. The target belongs to pure Gaussian chaos of degree $d = p+1$.

6.1 Two rounds and all-read noise

With exact measurements, read all r coordinates of U, then one unit projection of V along $c \odot \phi(U)$. This determines F_c exactly using $r+1$ reads.

If every read has independent variance-s Gaussian noise, let $\alpha = (1+s)^{-1}$. After $Y = U + \epsilon$, set $G = Y/\sqrt{1+s}$. Then G is standard Gaussian and

$$\mathbb{E}[\phi_\gamma(U) \mid Y] = \alpha^{p/2} \phi_\gamma(G).$$

Read V in the direction of this posterior coefficient. The explained variance is exactly $\alpha^{p+1}\sigma_c^2$, giving risk

$$R_{\text{two rounds}} = (1 - \alpha^{p+1})\sigma_c^2. \tag{7}$$

This equals the risk from observing every coordinate of (U,V) once with the same noise. Equality follows from the attenuation of a pure degree-$(p+1)$ chaos through the full Gaussian channel. It does not say that full coordinate observation is the optimal fixed design at its large budget.

6.2 A bound against every mixed fixed linear experiment

Theorem 3 (Universal fixed-design bound)**.** *For any fixed q unit-power linear reads of the joint (U,V), with independent noise variance $s \geq 0$, the explained variance is at most*

$$\min\left\{\sigma_c^2, \frac{(p+1)\,\|c\|_\infty^2\,\alpha q}{M}\right\}. \tag{8}$$

For $s = 0$ take $\alpha = 1$. The bound allows arbitrary mixed directions, repeats, oblique designs, independent randomized fixed designs and arbitrary nonlinear decoding.

Proof. Write $I_d(T)$ for the Wick polynomial of a symmetric order-d tensor T, normalized by $\mathbb{E} I_d(T)^2 = d! \|T\|_F^2$. For each Hermite feature choose a symmetric kernel A_γ in the U subspace with

$$I_p(A_\gamma) = \phi_\gamma(U), \qquad \langle A_\gamma, A_\eta \rangle = \mathbf{1}_{\gamma=\eta}/p!.$$

Let e_γ be the corresponding V coordinate, define

$$S = M^{-1/2} \sum_\gamma c_\gamma A_\gamma \otimes e_\gamma, \qquad T = \mathrm{Sym}(S).$$

Because the U and V subspaces are orthogonal, $F_c = I_d(T)$. For an exact fixed projection P, Gaussian conditioning projects every tensor slot, so explained variance is $d! \left\| P^{\otimes d} T \right\|^2$. Symmetrization is a contraction and commutes with the common slot projection. Dropping the first p contractions gives

$$\begin{aligned} d! \left\| P^{\otimes d} T \right\|^2 &\le d! \left\| P^{\otimes d} S \right\|^2 \\ &\le d! \left\| (I^{\otimes p} \otimes P) S \right\|^2 \\ &= \frac{d}{M} \sum_\gamma c_\gamma^2 \|P e_\gamma\|^2 \le \frac{d \|c\|_\infty^2 q}{M}. \end{aligned}$$

No step assumes P respects the U/V split. The conditional expectation is the optimal decoder under squared loss.

For noisy rows L, let $H = L^\top L$ and $P = H(H + sI)^{-1}$, the covariance of the posterior mean. Replace the slot projection by $P^{1/2}$. The same contraction argument bounds explained variance by $d \|c\|_\infty^2 \operatorname{tr}(P)/M$. If t_j are the eigenvalues of H, pad them to q entries. They sum to at most q. Concavity of $t/(s+t)$ gives $\operatorname{tr}(P) \le q/(1+s) = \alpha q$, proving (8). If q exceeds the ambient dimension, padding still gives a valid, possibly loose bound. □

The Gaussian Wick-conditioning identity also follows by conditioning the exponential generating function $\exp(t^\top X - \|t\|^2/2)$ and matching coefficients. The executable checks include independent cubic Gaussian quadrature, rather than only reusing the tensor implementation.

For fixed $p \ge 2$ and bounded nonvanishing coefficients, a constant-error fixed design needs $\Omega(M)$ reads, whereas the two-round construction needs $r + 1$. This is an accuracy separation with polynomially growing dimensions. The bound's constant is not asserted optimal.

7 Cubic features, finite context, and coefficient acquisition

For $p = 2$ the features are

$$\phi_{ii}(u) = \frac{u_i^2 - 1}{\sqrt{2}}, \qquad \phi_{ij}(u) = u_i u_j \quad (i < j), \qquad M = r(r+1)/2.$$

They obey

$$\|\phi(u) - \phi(v)\|^2 = \tfrac{1}{2} \left\| uu^\top - vv^\top \right\|_F^2. \tag{9}$$

Let $Q_{h,T}$ round each coordinate toward zero in increments h, clipping at magnitude T, with T/h integral. The message uses

$$B = r\,\lceil \log_2(2T/h + 1) \rceil$$

bits. Since this quantizer shrinks coordinate magnitudes, (9) and Cauchy–Schwarz imply

$$D_Q := \frac{1}{M}\mathbb{E}\,\|\phi(G) - \phi(QG)\|^2 \le 4h^2 + 4\mathbb{E}[Z^4 \mathbf{1}_{|Z|>T}], \quad Z \sim N(0,1). \tag{10}$$

For detail, writing $e = G - QG$ gives a bound $2\,\|G\|^2\,\|e\|^2$ on the feature error. Coordinate independence yields $4[\mathbb{E}Z^2 e_Z^2 + (r-1)\mathbb{E}e_Z^2]/(r+1)$ after division by M. Bound each rounding term by h^2 plus its Gaussian tail. The fourth tail moment is

$$2\{(T^3 + 3T)\varphi(T) + 3[1 - \Phi(T)]\}.$$

At $T = 6, h = 0.01$, the error is below 0.000411398 using 11 bits per context coordinate. This is an expected target-error bound, with no assertion that every coordinate simultaneously avoids clipping.

For known balanced coefficients, the quantized all-noisy protocol has exact risk

$$1 - \alpha^3 + \alpha^3 D_Q.$$

Both its direction and output amplitude use only the packed context. No precise analog first read is retained secretly by the decoder.

7.1 Learning coefficients in a declared layout

The context/response split and feature layout are supplied. The continuous coefficients c are unknown. Exactly M bounded synthetic function-value queries recover them: use $(U, V) = (0, e_{ii})$ for a diagonal coefficient and $(e_i + e_j, e_{ij})$ for an off-diagonal coefficient. The values are $-c_{ii}/\sqrt{2M}$ and $c_{ij}/\sqrt{M}$, respectively. Each full input has norm at most $\sqrt{3}$.

Fewer than M scalar queries cannot identify arbitrary c in an open coefficient domain. Every fixed adaptive transcript gives fewer than M linear constraints on c, leaving a nontrivial affine fiber. This is a coefficient-identification lower bound in this interface, not a bound on a narrower prediction objective.

If each calibration value has absolute error at most ν, then

$$\tau_c^2 := \frac{\|c - \widehat{c}\|^2}{M} \le (M + r)\nu^2. \tag{11}$$

The deployed true risk before final ADC and pattern rounding is exactly

$$(1 - \alpha^3)\sigma_c^2 + \frac{\alpha^3}{M}\mathbb{E}\,\|c \odot \phi(G) - \widehat{c} \odot \phi(QG)\|^2\,. \tag{12}$$

The second term is bounded using $[\tau_c + \|\widehat{c}\|_\infty \sqrt{D_Q}]^2$. Small adversarial calibration error is an explicit resource. A single physical read with deployment noise variance 0.01 does not acquire $\nu = 10^{-7}$ precision. Averaging, precision cost or a different calibration design would need to be charged in that alternative access model.

8 Finite output and finite local actuator weights

Let $A = c \odot \phi(G)$ and $\widehat{A} = \widehat{c} \odot \phi(QG)$. With ideal unit direction $w = \widehat{A}/\left\|\widehat{A}\right\|$, the last read is $Z = w^\top V + \xi$. Its law is $N(0, 1+s)$, independent of G. Let Q_z be a nearest-grid final ADC. The predictor is

$$\widehat{F} = \alpha^2 \left\|\widehat{A}\right\| Q_z(Z)/\sqrt{M}.$$

Set $k = \mathbb{E}[A^\top \widehat{A}]/M$, $t = \mathbb{E}\left\|\widehat{A}\right\|^2/M$, $a_z = \mathbb{E}[ZQ_z(Z)]$, and $b_z = \mathbb{E}[Q_z(Z)^2]$. Conditioning first on G, V and then on G, Z gives the exact identity

$$R_{\mathrm{ADC}} = \sigma_c^2 - 2\alpha^4 k a_z + \alpha^4 t b_z. \tag{13}$$

All four moments are computed from one-dimensional Gaussian bins. Monte Carlo is not used to establish this formula.

Clipping at 8 with step $h_z = 8/32767$ gives 65,535 values, fitting a signed 16-bit index. With $Z \sim N(0, 1+s)$,

$$\varepsilon_z := \mathbb{E}|Z - Q_z(Z)|^2 \le h_z^2/4 + \mathbb{E}[Z^2 \mathbf{1}_{|Z|>8}].$$

At $s = .01$ this upper bound is $1.490219 \cdot 10^{-8}$. The implementation actually serializes and decodes these indices.

8.1 The dense program is finite and remains paid

A compact remote descriptor does not mean a small local actuator. To make this explicit, round each ideal w_i to $j_i/2^{18}$. Let $k_M = \lceil \sqrt{M} \rceil$, $D = 2^{19} + k_M$, and implement

$$\widetilde{w}_i = 2j_i/D.$$

Each $j_i \in [-2^{18}, 2^{18}]$ fits a 20-bit alphabet. Since $\|2j - 2^{19}w\| \le \sqrt{M}$,

$$\|\widetilde{w}\| \le 1, \qquad \|\widetilde{w} - w\| \le \frac{\sqrt{M} + k_M}{2^{19} + k_M} =: \delta_w.$$

The actual pattern belongs to the unit ball, not necessarily the unit sphere. This is consistent with the power cap and with the stronger unit-row fixed comparator.

Coupling the same V and read noise bounds the output's L^2 change by $\alpha^2\sqrt{t}\,\delta_w$. After rounding, the last read has conditional variance $\|\widetilde{w}\|^2 + s \le 1 + s$. Its distribution need not be independent of G, but the ADC bound holds uniformly. Thus

$$R_{\mathrm{actual}} \le \left[\sqrt{R_{\mathrm{ideal,before\ ADC}}} + \alpha^2\sqrt{t}\,(\delta_w + \sqrt{\varepsilon_z})\right]^2. \tag{14}$$

The exact unit-pattern identity (13) must not be reused after this contraction. The executable packs and decodes every 20-bit pattern index and checks $4\sum_i j_i^2 \le D^2$ with integer arithmetic, verifying the rational power cap. An accompanying proof supplies exact integer comparisons to generate indices near rounding thresholds from rational prepared coefficients; the executed dense generator uses float64.

9 A final-accuracy comparison with all resource counts visible

Consider the separate normalized case $r = 128$, $M = 8256$, $c = 1$, and $s = .01$. The target variance is one. All guarantees below refer to absolute squared error in these units.

Experiment or bound	Reads	MSE
Two-round protocol, ideal context and pattern	129	0.0294098521
Packed context and final ADC, ideal unit pattern	129	0.0295043397
Packed context, 20-bit local pattern, final ADC	129	≤ 0.0299713103
Every fixed linear design at the same budget	129	≥ 0.9535891089
Full coordinates, each observed once noisily	8384	0.0294098521

The first and full-coordinate rows are exact ideal risks. The second is an exact Gaussian-bin calculation. The third is a conservative theorem bound. The fourth is a universal lower bound, not an experimentally optimized baseline. All fixed rows allow arbitrary joint-coordinate directions and arbitrary nonlinear decoding.

The fixed bound implies that MSE at most 0.03 requires at least

$$\left\lceil \frac{0.97M}{3\alpha} \right\rceil = 2697$$

fixed reads. This is *necessary*; it is not asserted sufficient. Reading all 8384 coordinates supplies a sufficient comparison with a larger budget.

The constructive protocol uses 1408 first-ADC bits, a 1408-bit remote control packet, a 16-bit final readout, and a dense local program of 165,120 bits (20,640 bytes). It computes and applies 8256 coefficients. The local program size is this construction's encoding upper bound, not a lower bound on all possible actuators. If the sensor retains the first record locally, the separate remote packet need not travel, but the local work remains.

9.1 Executed acquisition and deployment

The acquired case has $r = 64$, $M = 2080$, and hidden continuous coefficients in $[0.9, 1.1]$. It is distinct from the normalized $r = 128$ theorem example. The final fixed comparator is given the *true* coefficients, stronger than an equally calibrated competitor.

Resource or result	Acquired case
Scalar calibration calls	2080
Calibration feature entries materialized	4,326,400
Calibration dense target products	8,652,800
Calibration absolute value error budget	10^{-7}
Prepared model storage	16,640 bytes
Online scalar reads	65
First ADC / remote control packet	88 / 88 bytes
Local finite pattern / final ADC	5200 / 2 bytes
Actuator weights / dense projection products	2080 / 2080
Fresh fully finite-pattern trials	8192
Measured fully finite-pattern MSE	0.029191553
Estimated standard error of that mean	0.000474301
Proved finite-pattern MSE upper bound	0.029789756
Risk of every fixed 65-read design	≥ 0.893996249
Full 2144-coordinate noisy risk	0.029595270

The separate ideal-unit-pattern, finite-ADC run predicts MSE 0.0296898866 exactly and measures 0.0299856295 with estimated standard error 0.0004821124 on 8192 fresh trials. Agreement is about 0.62 estimated standard errors. The finite-pattern run uses a different seed; its lower observed mean is not evidence that rounding improves risk. The sample uncertainty is reported, rather than treating Monte Carlo as an exact optimum.

Calibration is executed through the function-value interface. Hidden coefficients and full input vectors are available only to the calibration evaluator and independent scorer. Replaying calibration to verify the retained model incurs additional validation calls; it does not create new scientific information. Deployment need not repeat that verification. Preparation, scoring and verification costs are not used to support a runtime advantage.

9.2 Strong ordinary alternatives

If V is stored digitally, directly evaluating $c \odot \phi(U)$ against it takes $O(M)$ arithmetic. Generating the adaptive pattern and performing its dense projection also takes $O(M)$. If implementing the scalar aggregate first requires exporting all of V, the proposed acquisition saving disappears.

For the random function $f_V(u) = M^{-1/2}(c \odot \phi(u))^\top V$, a native point-value query at $u = U$ is the ordinary ideal final operation. A direct-target or suitable quadratic-energy interface therefore ties it, subject to its actual normalization and noise contract. This construction is not a faster dense polynomial evaluator. The meaningful access question is whether the source can implement the aggregate locally from the compact descriptor at lower total cost than the best available ordinary alternative. No such apparatus is demonstrated here.

9.3 Robustness does not certify model membership

The separation is stable under a specified approximation error. Let the actual target be $F = F_0 + e$ on the same input distribution and sensing interface, with $\|e\|_{L^2} \leq \delta$. If a predictor has model risk at most R_0 and every fixed experiment has model risk at least L_0, the triangle inequality gives

$$R_{\text{actual}} \leq (\sqrt{R_0} + \delta)^2, \qquad R_{\text{fixed,actual}} \geq (\sqrt{L_0} - \delta)_+^2.$$

These inequalities extend the structural comparison to a neighborhood of the declared family. They do not acquire δ.

A companion provides a paid statistical route when fresh full-input target evaluations and a valid residual moment bound $\mathbb{E}e^4 \le K(\mathbb{E}e^2)^2$ are supplied. Use blocks of $b = \lceil 16K \rceil$ independent squared residuals and the smallest odd number $t \ge 8\log(1/\beta)$ of blocks. Chebyshev's inequality makes a block's relative error at most $1/2$ with probability at least $3/4$; a majority bound gives the median M_e the same accuracy with probability at least $1-\beta$. Then $\mathbb{E}e^2 \le 2M_e$.

This certificate costs bt fresh full-input samples, bt target evaluations, bt reference evaluations and their full input traffic. A numerical residual error at most τ gives the upper L^2 certificate $\sqrt{2}(\sqrt{\widetilde{M}_e} + \tau)$. The moment bound and sampling contract are substantive assumptions; they are not established by passing a finite holdout. No such extra target-membership experiment was executed for an external scientific system.

10 The representation-discovery boundary

The constructive family declares its split and Hermite layout. This is not a superficial convenience that can be replaced by an unspecified efficient discovery algorithm.

Theorem 4 (A cubic recognition obstruction)**.** *Given an explicit homogeneous cubic P with integer coefficients, deciding whether there is a nonzero direction d for which $t \mapsto P(x+td)$ is affine for every x is NP-hard. The reduction has a polynomial promise gap: YES instances admit unit d with $\|\nabla P(d)\| = 0$; NO instances satisfy*

$$\|\nabla P(d)\| > \frac{1}{384N} \quad \textit{for every unit } d,$$

where N counts Boolean and auxiliary variables in the reduction.

Proof. For a 3-SAT formula with n variables and m clauses, introduce a homogenizing variable t and one Boolean auxiliary a_j per clause. Let $N = n + m$. Every Boolean or auxiliary variable y has constraint $y^2 - ty = 0$. A positive literal has false-indicator $\ell = t - x_i$ and a negative literal has $\ell = x_i$. For each clause add

$$\ell_1\ell_2 - ta_j = 0, \qquad a_j\ell_3 = 0.$$

Call these homogeneous quadratics $q_j(v)$. If $t = 0$, every coordinate is zero. Otherwise division by t gives Boolean variables and exactly the satisfying-assignment constraints. Hence they have a common nonzero real zero precisely when the formula is satisfiable.

Introduce a scalar w and a vector z indexed by the quadratic constraints, and define

$$P(w, z, v) = w^3 + 3w\,\|z\|^2 + \sum_j z_j q_j(v).$$

At $d = (a, b, h)$ the w derivative is $3(a^2 + \|b\|^2)$. Thus a gradient zero forces $a = b = 0$, and the remaining z equations become $q_j(h) = 0$. For a homogeneous cubic,

$$D_d^2 P(x) = 2\nabla P(d)^\top x,$$

so gradient zeros are exactly the global affine directions.

For the gap, suppose $\|v\|^2 \geq 1/2$ and every quadratic residual is at most $\varepsilon \leq 1/(192N)$. From $|y_i^2 - ty_i| \leq \varepsilon$ follows $|y_i| \leq |t| + \sqrt{\varepsilon}$ and

$$1/2 \leq (2N+1)t^2 + 2N\varepsilon, \qquad t^2 \geq 1/(12N).$$

After division by t, the normalized residual is at most $\delta = \varepsilon/t^2 \leq 1/16$. Each variable lies within 2δ of 0 or 1. Rounding gives an integer gate residual of magnitude at most $8\delta \leq 1/2$ and an integer clause residual of magnitude at most $6\delta < 1$, so both vanish. The formula is satisfiable.

Now if a unit d has $\|\nabla P(d)\| \leq \eta \leq 1/(384N)$, its w component gives $a^2 + \|b\|^2 \leq \eta/3$, hence $\|h\|^2 \geq 1/2$. Each z component gives

$$|q_j(h)| \leq \eta + 6|ab_j| \leq 2\eta \leq 1/(192N).$$

The previous rounding argument then implies satisfiability. Contraposition proves the gap. Dimensions and coefficient bit lengths are polynomial. □

For Gaussian base point X, the expected squared directional curvature is $4\,\|\nabla P(d)\|^2$, so the gap is not restricted to an exact equality at inaccessible precision. Polynomial normalization preserves an inverse-polynomial gap. The result concerns global affine-direction recognition. It is not a hardness theorem for all adaptive policies, all approximation criteria, or an optimal sensing budget. A single affine direction need not save a total read. Broad tensor-hardness results are established context [4]; this scoped reduction is proved directly.

11 Failed transfers and useful companions

11.1 A physical quadratic-form interface was examined, not assumed

Astra's configured-shunt construction supplies a conditional model

$$y = b + \frac{\lambda}{1 + \lambda v^\top H v}, \qquad v^\top H v = \frac{1}{y-b} - \frac{1}{\lambda}.$$

Its governing source expressly requires a genuinely configurable rank-one shunt and does not demonstrate that hardware [8]. Its bounded positive-definite source law is not automatically compatible with the unbounded Gaussian matrix law in the Hermite example. Moreover, a native quadratic-energy read is already a task-specific direct-target alternative. The source and its scoped corrections were read; this transfer was not credited as an acquired physical capability.

11.2 Posterior simplification closes on a promised class, but ordinary methods tie

An independent constructive attack studied

$$V(x) = \tfrac{1}{2} x^\top H x - \sum_j a_j \log\cosh(v_j^\top x), \quad H \succ 0,\ a_j \geq 0,$$

with zero hidden biases. Here $H_0 = \nabla^2 V(0)$ is a global lower Hessian bound. Gaussian measurement precision R makes every posterior η-strongly convex exactly when $H_0 + R \succeq \eta I$.

For unit-power reads with noise variance σ^2, writing $D = (\eta I - H_0)_+$, the exact minimum count is

$$\max\{\operatorname{rank} D, \lceil \sigma^2 \operatorname{tr} D \rceil\},$$

with zero reads if $D = 0$. A classical frame-completion construction attains it [6]. A value-only calibration acquires a guarded certificate in $n^2 + n + 1$ scalar potential evaluations under supplied fourth-derivative and precision bounds. The executed six-dimensional example uses 43 calibration values and three reads; a uniform-coordinate sufficient design uses twelve. Equally informed ordinary spectral design ties the construction exactly.

The companion develops value-only gradient costs and a conservative normalization closure. Its elementary bound asks for roughly $3.68 \cdot 10^{28}$ scalar values even on the two-dimensional toy and was not executed. Established log-concave normalization methods are stronger machinery [7]. The source prior normalizer also remains unknown: conditioning on a real observation does not compute it for free. The complete proof, sources, measured kernel operations and this failure are retained in the evidence archive.

12 Validation, preservation and claim status

The main Gaussian fixed-design proof was independently reconstructed. Executable checks passed 102 mixed noiseless designs, 306 mixed noisy designs and 27 independent cubic Gaussian quadratures. The cubic hardness executable checked 2048 assignment encodings, 1024 satisfying directions and 256 exact derivative identities, and preserves a 40-dimensional unsatisfiable cubic with 88 monomials and coefficients bounded by 3.

Two independent audits verified the noisy replay theorem and its access conditions. A 100,000-draw interleaved noisy example matched its posterior-variance identity within 0.052 paired standard errors. A nonlinear first-stage example tested the averaged J_0 decomposition while retaining nonzero conditional covariance terms. These checks support the ordinary proofs; none is used as a substitute for them.

The previous policy-independent residual-risk theorem, sharper quadratic bound, rare-concentration obstruction, acquisition constructions, two-round comparison, noisy optimal-fixed separation, inertia classification, coordinate companion, acquired implementations, corrections and useful failures remain in their original artifacts. The new work is isolated in a separate directory. The included preservation check compares previous declared SHA-256 manifests without rewriting their files.

Claim	Status
Noisy interactive-to-codebook value identity	Proved; two independent audits; hard budgets and explicit decoder access
Covariance insufficiency for controlled experiments	Proved by equal-second-moment sources with different description geometry
Hermite fixed-versus-two-round accuracy separation	Proved for all mixed fixed linear designs; not an exact fixed optimum
Finite messages and finite local pattern	Constructed, packed and executed in software; local size and work charged
Coefficient acquisition	Executed in declared layout with supplied precision contract
Generic layout discovery	Not provided; a related cubic recognition task is NP-hard
Physical sensor / full-cost speed advantage	Not established
Historical novelty	Unresolved; classical machinery and close prior work explicitly credited

The strongest completed consequence is an exact organizing theory for a conditional problem: early information changes the distribution of the remaining task description, and feedback value is the distortion of that description under feasible experiments. The constructive separation survives noisy reads and finite messages. The next decisive operational gate remains a source that really supplies the required aggregate, with acquired model adequacy and a lower total cost than its strongest direct-target and full-input alternatives.

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